

Data Explorations

This dataset is sourced from Statistics Canada (StatCan). The original title of the dataset is **Monthly retail trade sales by province and territory (x 1,000)**.

StatCan allows public to download data in different forms and sizes. The dataset is obtained in its entirety as CSV file and is explored using Transform data tab in Power BI Desktop. It spans over the period of nine and a half years (January 2017 to June 2026). It consists of 79343 rows and 17 column in its raw form. There are more regions included in this dataset than just the number of provinces and territories in Canada. The examples of such regions are Toronto, Ontario and Edmonton, Alberta and Montreal, Quebec. These regions are simply the sub-regions of actual regions. All the actual regions are also the sub-regions of Canada, which is also shown as single region. This makes the total number of regions and sub-regions equal to 23. However, the measures and other calculations will be applied only on actual regions. The units used for the amount from trade sales are Dollars and the associated values are scaled down by 1000 times as given in the title. The data quality levels are also labelled using alphabet A to F, A being the most accurate or of excellent quality.

There are about 30 industry categories and two modes of Sales, namely Total Retail Sales and Retail e-commerce sales; former being over 99% of all of the sales. Data form also has two categories seasonally adjusted and unadjusted. 13 % of sales Values are null and the corresponding cells to null values are labelled x in the Status column. Column profiling option in Power BI also shows that about 9000 cells have x. This means that out of 13% (10319 data points) unavailable data, 8980 data points are suppressed due to confidentiality reasons.

Details of other StatCan units and columns which are more important only for internal references are given in a supporting document named 'STATCAN Format Columns'. Sales column has two categories: Total, retail sales and Retail E-commerce sales where E-commerce are less than 1 %.

Data Cleaning

The columns explained above are only 7. The rest of the columns in the dataset are not mentioned above as they were removed from the dataset. The removed columns are useful to understand a grain of data properly, however, they are not desired in dashboard for analytic purposes. The seven columns were renamed to make them more understandable in simple language. Renaming is explained as follows:

Original name	New name
Ref_Date	Date
Geo	Region
North American Industry Classification System (NAICS)	Industry
Sales	Sales
Adjustment	Data form
Value	Amount 1000 \$
Status	Status

Columns like Data form, Status, and Sales are not renamed or deleted because these columns are not directly contributing in building dashboard, however, they will be always important for quick references.

Data Preparation

All the data types were left as they were. The date column namely Dated is the only of type Date whereas rest of the columns are of text type. To avoid unnecessary confusion and potential mistakes of double counting, the sub-regions of provinces were filtered out during the transformation stage. The provinces and territories plus Canada was kept, in case, if National versus Provinces comparisons were to be made. The sketch of dashboard is planned based on the questions to be answered. The questions, the thought process and the intended choice of visuals in the dashboard are as follows:

1. How have total retail sales trended monthly/yearly from 2017 to mid-2026, and is there a visible seasonal pattern?

Line graph of Amount x 1000 \$ (retail sales values) over Year and Month.

2. How does e-commerce retail sales compare to total retail sales over time — is e-commerce's share growing?

(Although, not completely clear on this question but I think) A vertical stacked bar chart of retail sales over Dated [Year] and [Month] may show the comparison between total retail sales, e-commerce retail sales and over all retail sales, especially with sales units switched to %. However, including monthly details beside [Year], chart becomes messy and less obvious to display the differences. Therefore, using only Dated [Year] and [Month] displays the answer more evidently.

3. Which provinces/regions have the highest and lowest retail sales, and how does that ranking hold up over time?

A horizontal bar chart over regions would reveal the highest and lowest retail sales in a ranked manner. Use of Year slicer can displays the ranking over time.

4. Are there specific periods (e.g., 2020) where sales patterns broke from the norm, and how visible is that in adjusted vs. unadjusted data?

A slicer of adjusted and unadjusted categories of sales would be helpful here. When filters in slicer applied, the pattern changes against the norm will be revealed on the existing line chart of Amount x 1000 \$ over versus time (year and month).

5. What's the most recent month's performance compared to the same month a year prior?

Key performance indicators (KPIs) of Month over Month (MoM) and Year over Year (YoY) would display MoM and YoY comparisons.

Besides these KPIs, an Industry slicer, Sales slicer and other KPIs like overall retail sales, number of industries would be invaluable additions to the dashboard.

Data Modelling

A few measures have been calculated. A couple are given as follows:

The Total Retail Amount

```
Total Retail Amount = CALCULATE (
SUM ('Retail Sales Canada'[Amount x 1000 $]),
'Retail Sales Canada'[Region] = "Canada",
'Retail Sales Canada'[Sales] = "Total retail sales",
'Retail Sales Canada'[Data Form] = "Unadjusted")
```

The filters are included to avoid double or triple counting of regions, retail sales and adjusted/unadjusted data. This measure is used to display KPI of Total Retail Amount.

Sales_Total Retail Amount:

```
Sales_Total Retail Amount =
CALCULATE (
SUM ('Retail Sales Canada'[Amount x 1000 $]),
'Retail Sales Canada'[Region] = "Canada")
```

In this case, the sales filter is removed because it is visual 1 where both sales (total retail and e-commerce) are needed to be displayed.

Region_Total Retail Amount

```
Region_Total Retail Amount =
CALCULATE (
SUM ('Retail Sales Canada'[Amount x 1000 $]),
'Retail Sales Canada'[Sales] = "Total retail sales",
'Retail Sales Canada'[Data Form] = "Unadjusted",
KEEPFILTERS ( 'Retail Sales Canada'[Region] <> "Canada" )-- filtering out
Canada)
```

Here, only Canada is filtered out among all other filters, because in Visual 3, Canada, being not the true region within Canada, was wrongfully adding to actual sum of regional sales.

The other measures were calculated are previous year, previous month, YoY and MoM. No additional columns or measures were needed. The data available in the current form with measures needed suggested that all the requirements to build the desired dashboard.

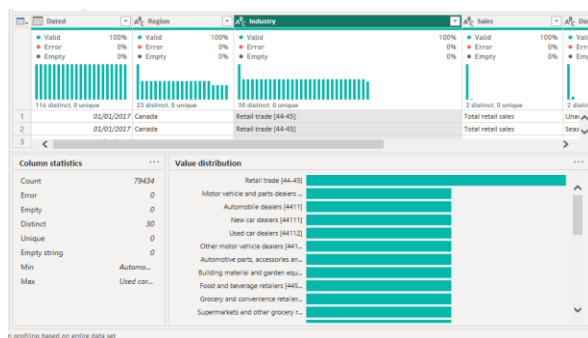
Dashboard Building and its Components

The decided components of the Dashboard are as follows:

- Visual 1: A line chart of the Sales_Total Retail Amount versus the Year and Month. The subtitles and legend displays details of calculations employed.
- Visual 2: A vertical clustered bar chart of Sales Amount by Categories over years.
- Visual 3: A horizontal bar chart of the regions over the Region_Total Retail Amount.
- Four slicers including Year, Industry, Data Form and Sales.
- 3 KPIs: Total Retail Amount, Year over Year and Month over Month.

Dashboard Snaps

Some snaps from the dashboard and work process are shown here. The top left depicts the Transform tab with column profile which helps to easily view detailed information of a table. The top right is the snap of main dashboard without any selection of highlighted component. The middle left shows metrics and visuals of 2025 only, selected from the year slicer. The middle right is the snap taken when seasonally adjusted data is sliced from Data form slicer. The overall trend only shows April 2020 tip while diminishing all seasonal dips in every year. The bottom left all metrics for only Cannabis industry whereas the bottom right shows visuals and metrics based only on beer, wine and liquor industry.



Questions Answered

1. How have total retail sales trended monthly/yearly from 2017 to mid-2026, and is there a visible seasonal pattern?

Visual 1 displays the total retail sales have been gradually rising for the years from 2017 to mid-2026. The overall retail sales has reached to 21.05 billion dollars till June 2026. The sales by

individual years can be seen in Total Retail Sales KPI by applying yearly selections in the year slicer. For example, in 2017, the total retail sales were equal to 1.87 billion, then it kept gradually increasing till the end of June 2026.

However, the monthly trend of total retail sales is almost the same each year of the period.

Every year, the trend consistently remains the same with an exception of April 2020.

Generally, the sales begins slowly from January, then gradually rises in February. After that from March till December, the highs and lows become very smooth.

2. How does e-commerce retail sales compare to total retail sales over time — is e-commerce's share growing?

It can be seen in visual 2 that e-commerce retails are negligible compared to total retail sales over all the months. They differ roughly by multi-folds throughout this period. Using Sales slicer, it can also be seen that, 2025 has the highest e-commerce sales which barely reaches 119 million. On the other hand, the lowest e-commerce sales are in 2017 that is under 32 million. It, however, has a rising trend over the years. The overall trend of both categories of sales can be considered the same, however, the scale of retail sales amount are dramatically different.

3. Which provinces/regions have the highest and the lowest retail sales, and how does that ranking hold up over time?

The ranking is established over all Canadian provinces and territories in visual 3. The highest and the lowest retail sales have been very consistent over the given period. Ontario, Quebec and British Colombia are consistently the top 1, 2 and 3 regions respectively. The overall Ontario's sales reach about 8 billion whereas Nunavut represents the lowest rank with 13 million of sales.

4. Are there specific periods (e.g., 2020) where sales patterns broke from the norm, and how visible is that in adjusted vs. unadjusted data?

As mentioned above, every January and February are slower than the other months which are seasonal trends as they can be seen consistently throughout the period. However, April 2020 is the most unusual dip in the pattern. With the use of Data form slicer, seasonally adjusted filter displays the overall trend more smoothly. Under this setting, the seasonal dips of January and February are diminished resulting in the even more prominent unusual dip around April 2020.

5. What's the most recent month's performance compared to the same month a year prior?

The most recent month would be June of 2026. The comparison of the performances of the most recent month and the same month in the prior year can be displayed Total Retail Amount by selecting specific months. In the most recent month, the retail sales amount is 235.94 million whereas the retails sales amount in the same month prior year is 209.87 million. This reveals 6.22% YoY growth from June 2025 to June 2026.

Additional Insights

The dashboard is built more than just displaying answers to the questions asked. An

important visual that has not been discussed yet is Industry slicer. There are about 30 different industries contributing in this retail record. Single industry based insights can be revealed using the industry slicer. For example, when Beer, wine and liquor industry is selected, valuable information is revealed. The dashboard shows that overall 240.6 million retail sales have been made, annually it has grown by 4.06% and British Colombia beats Quebec in ranking, however, Ontario still holds the rank 1. The overall trend of liquor industry over time is also very similar to that of all the industry together.

In case of Cannabis industry, the overall trend shows much more prominent rise from 2017 till 2026. Alberta beats both British Colombia and Quebec here and becomes second highest retail sales amount generator.

Dashboard is free to use to explore more insights of the dataset from industry viewpoint or any other perspective.